Professor Quant Partnership Model

A simulation of an LLM-enabled funding model for university research labs

Repository: https://github.com/huazuhao/professor_quant_partnership_model

This repository studies a new research-lab funding model made possible by the rapid improvement of LLM coding.

University labs in mathematics, statistics, optimization, simulation, computational modeling, and related STEM fields already have the talent systematic hedge funds rely on. The old bottleneck was implementation: turning a research idea into clean data, backtests, diagnostics, risk controls, deployment code, and monitoring usually took more engineering time than a short sabbatical or research leave could support.

LLM-assisted coding changes that. A 6-12 month sabbatical or one-year leave can now be enough for professors to help invent or improve systematic trading strategies. If validated strategies are traded, part of the profits can fund the contributing labs over the following years.

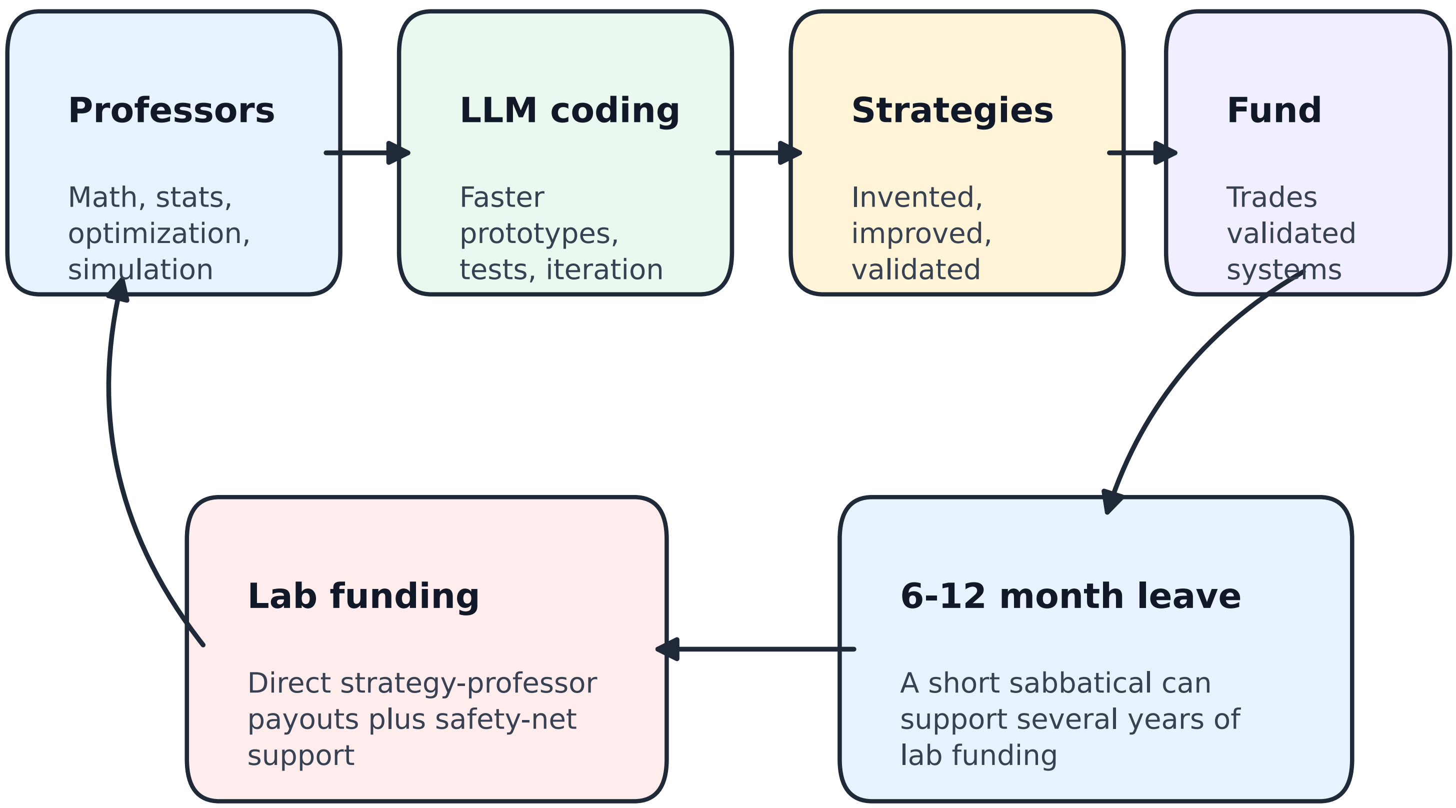
In the current simulation, this loop can start with $20M of initial capital and grow into a $1B+ fund. The median 10-year investor outcome is about 3x capital, roughly 140-150 professors or labs participate, and five-year lab compensation is above $1M at the median.

What the simulation tests

Whether focused professor research, LLM-accelerated implementation, fund infrastructure, strategy profits, and recurring lab support can become a large alternative funding channel for universities.

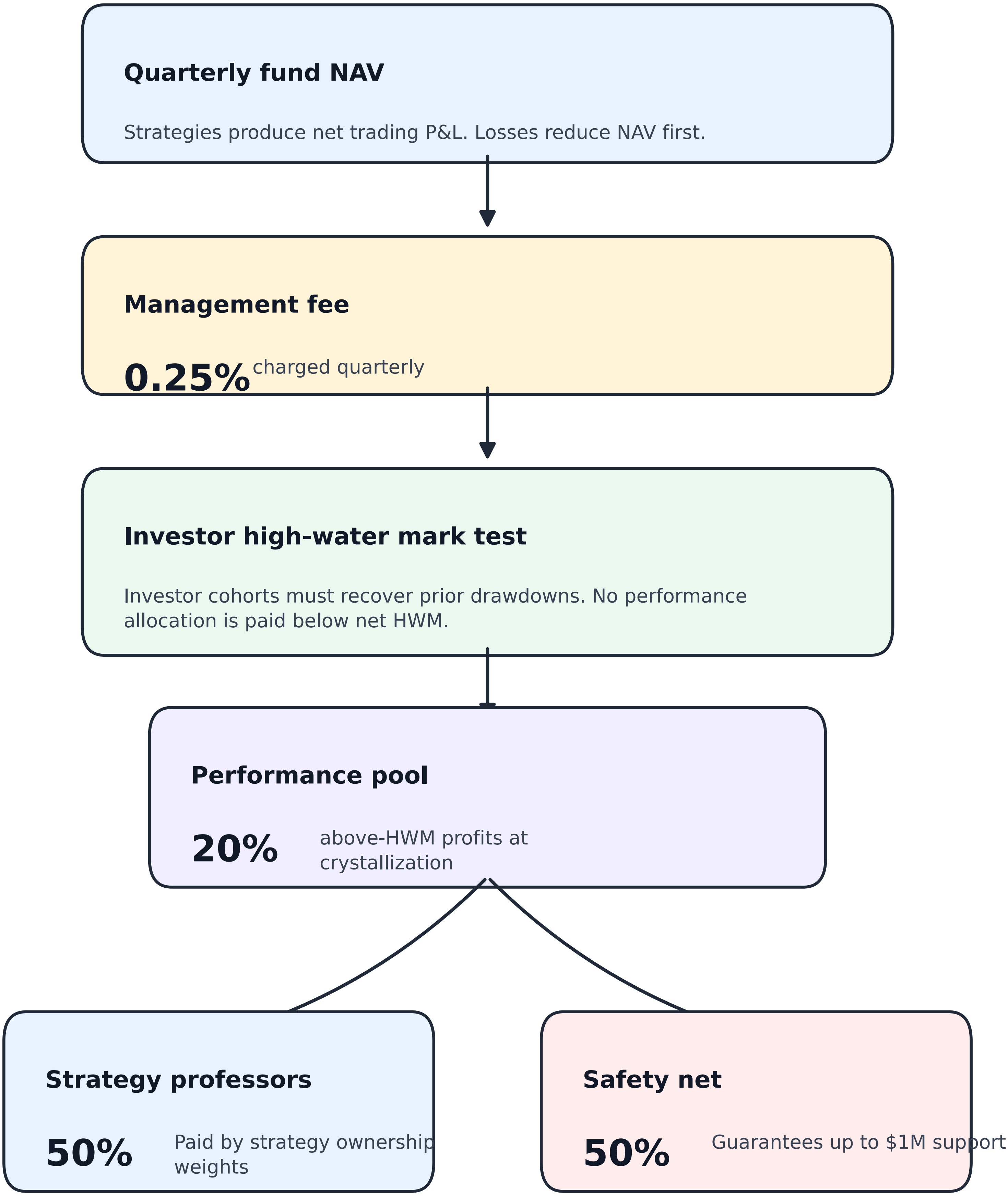
The Funding Loop

A short research leave becomes strategy development, fund infrastructure, and recurring lab support.



Current Simulated Accounting

Performance compensation is paid only after investor drawdowns are recovered.



Eligible profits after HWM: 80% fund/investors, 10% strategy professors, 10% safety net.

Current Simulation Results

Baseline batch run: 100 paths, 40 quarters, seed 42.

Metric	Current baseline result
Initial capital	**$20M**
Final fund AUM after 10 years	**about $1.27B median**
10-year investor outcome	**about 3x capital**
Professors/labs after 10 years	**138.5 median**
Five-year professor/lab compensation	**$1.34M median**
Five-year compensation, p75	**$1.63M**
Five-year compensation, p90	**$2.00M**
$100 reference investment ending value	**$302.52 median**

Interpretation

These are simulation outputs, not forecasts. The purpose is to test whether the economic loop is internally coherent and whether the scale of the research-funding impact is large enough to justify deeper investigation.

Chart 1: Investor Outcome Distribution

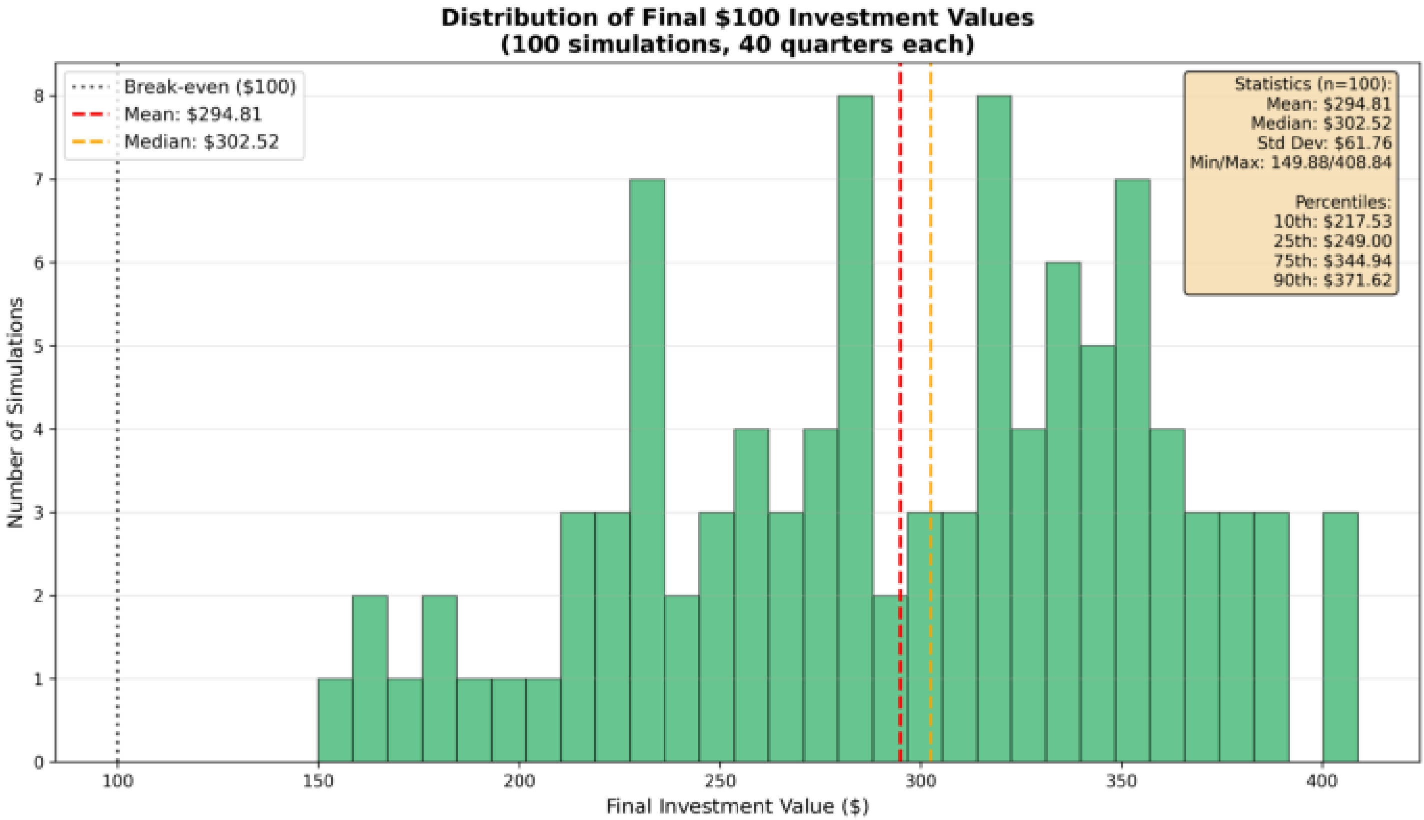


Chart 2: Five-Year Professor/Lab Compensation

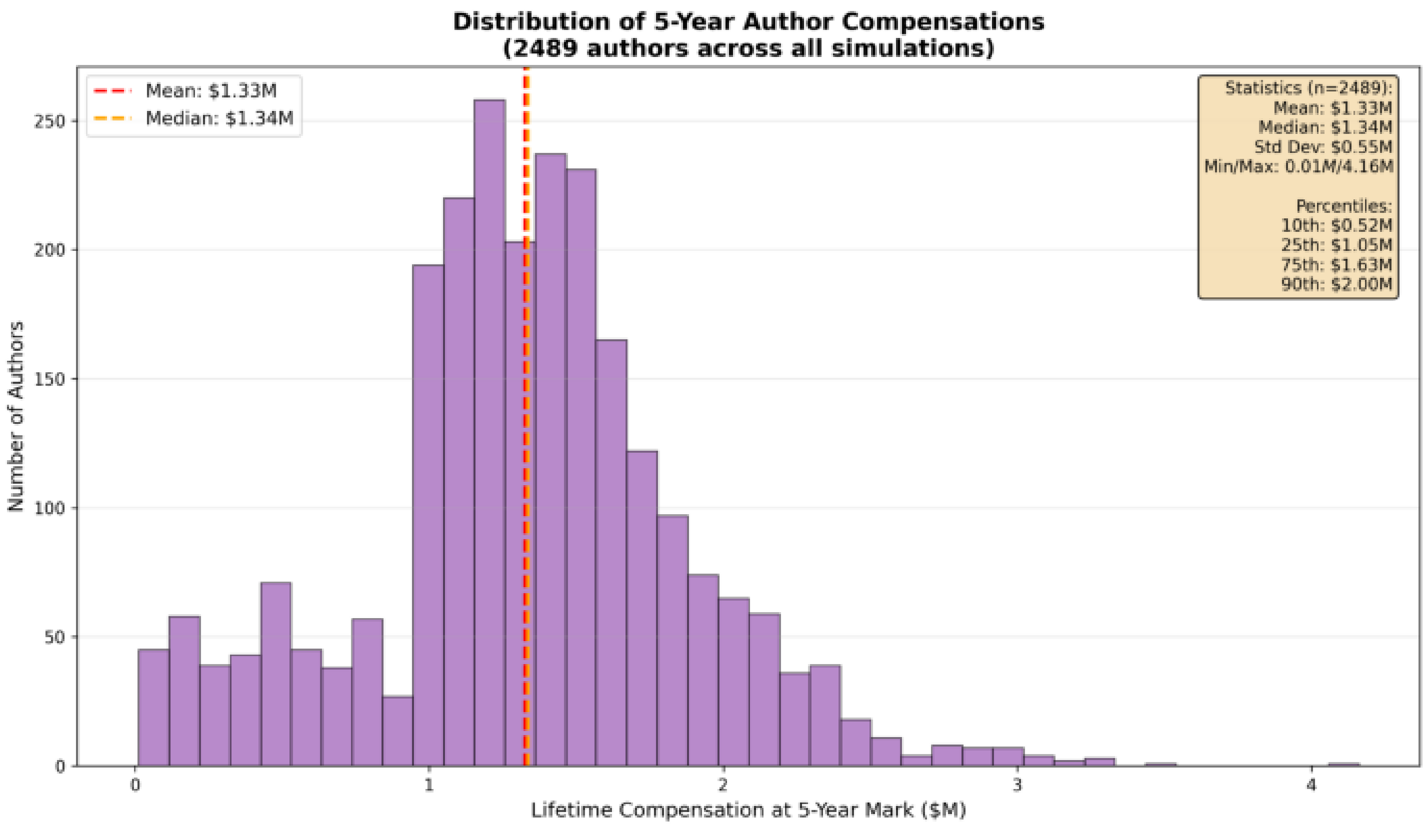


Chart 3: Final Fund AUM

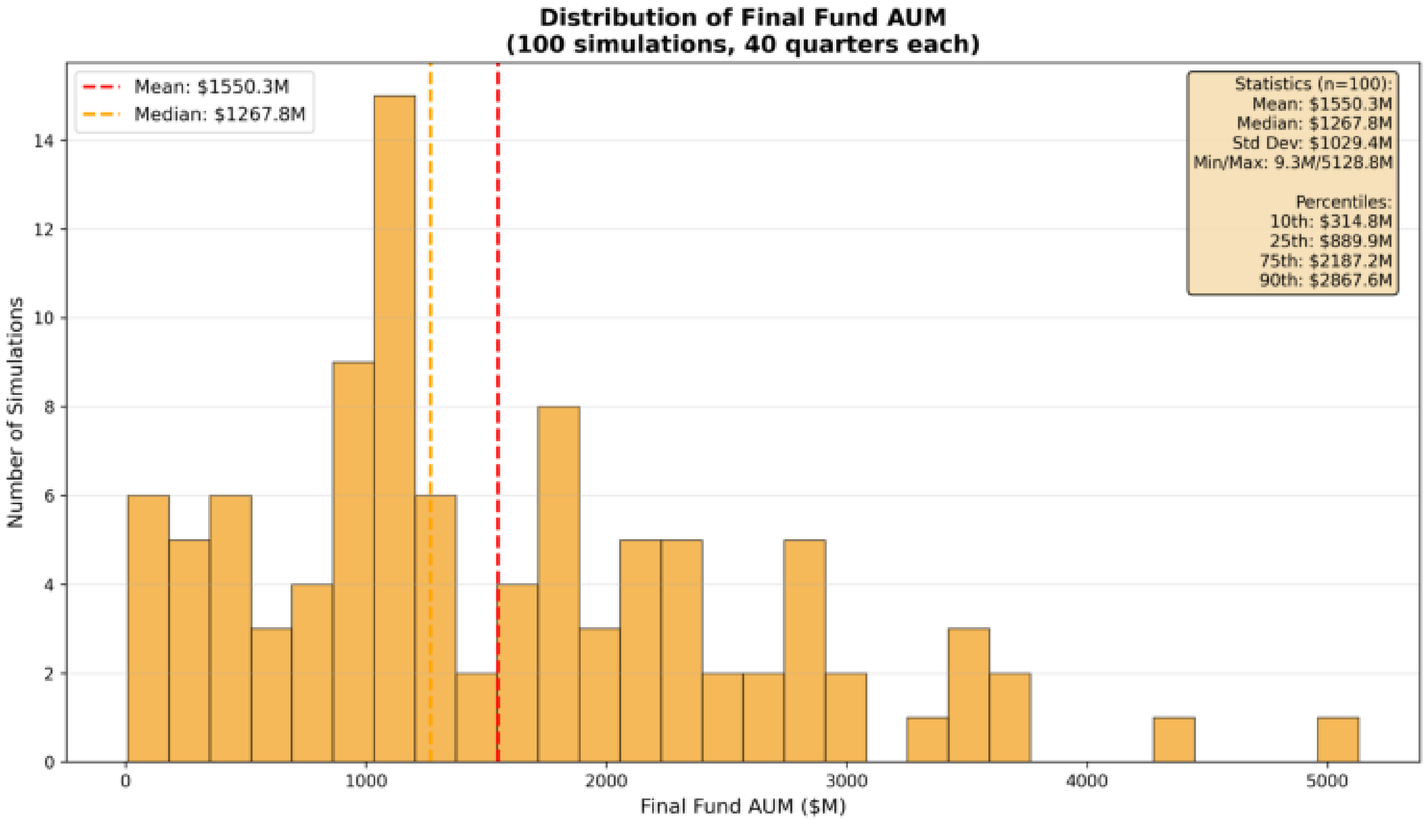


Chart 4: Total Professors/Labs

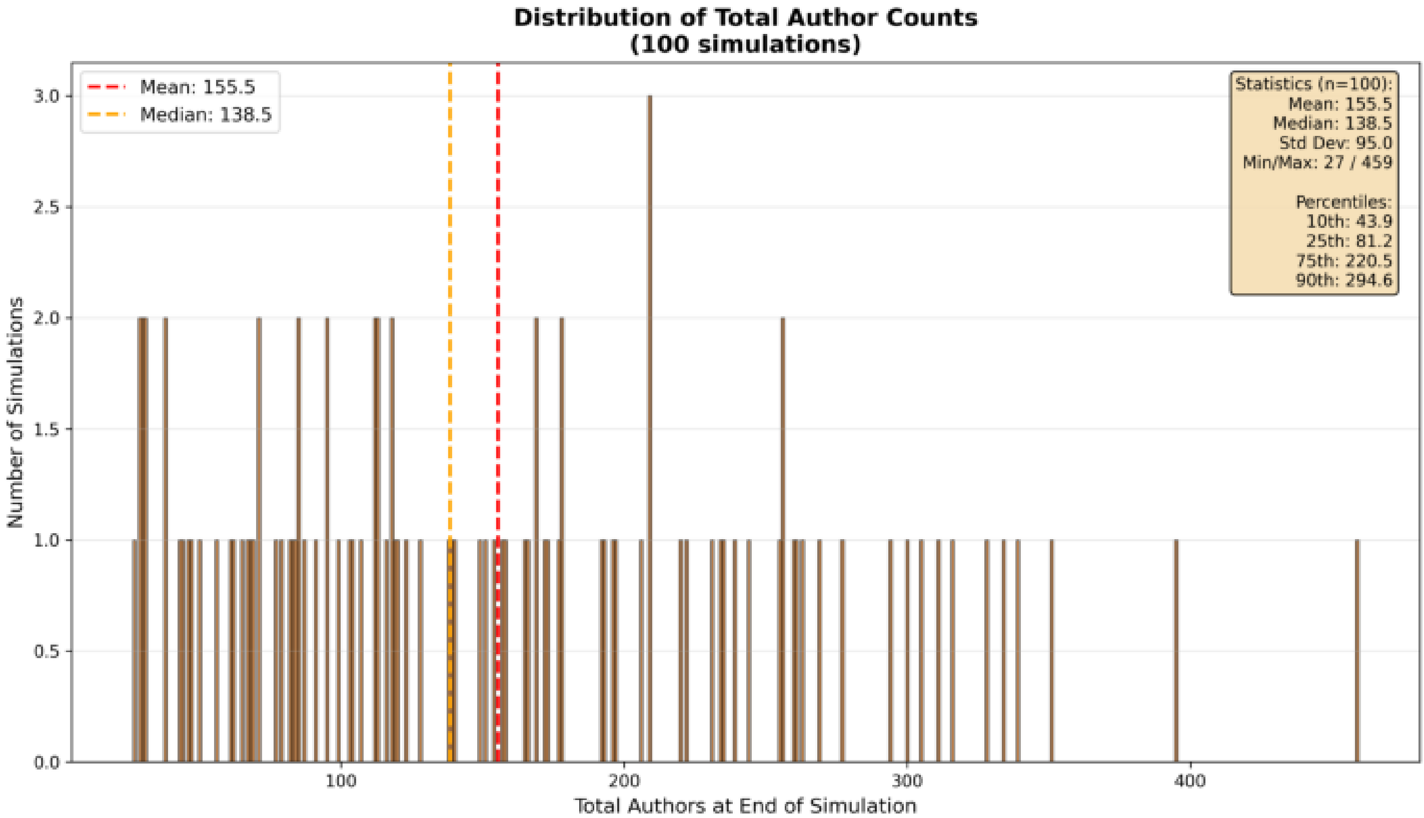


Chart 5: Paid Professors/Labs